



COMSATS UNIVERSITY ISLAMABAD

Lahore Campus: Defence Road, Off Raiwind Road, Lahore

UAN: +92-42-111-001-007 Ext: 810, 866

Assignment#1- Semester Fall 2025

Course Title:	Digital Signal Processing			Course Code:	CPE 324	Credit Hours:	4(3,1)
Course Instructor/s:	Dr. Arsla Khan			Programme Name:	BCE		
Semester:	5 th	Batch:	FA23-BCE	Section:	A+B	Given Date:	12-10-2025
Submission Date:	19-10-2025 (12:00 PM)			Maximum Marks:			10
Student's Name:				Reg. No.			

Important Instructions / Guidelines:

- Write your **Name** and **Student registration No.** clearly on the first page.
- Please properly label x-axis and y-axis of your **graphs**.
- Provide **all details** to get full credit.

Question 1: Determine the system function of a discrete-time system described by the difference equation

$$y(n) - \frac{1}{3}y(n-1) + \frac{1}{5}y(n-2) = x(n) - 2x(n-1)$$

Question 2: A causal system has input $x(n]$ and output $y(n]$. Find the system function , frequency response and impulse response of the system if

$$x(n) = \delta(n) + \frac{1}{6}\delta(n-1) - \frac{1}{6}\delta(n-2)$$

and

$$y(n) = \delta(n) - \frac{2}{3}\delta(n-1)$$

Also assess the stability of the system.

Question 3: A causal system is represented by

$$H(z) = \frac{z+2}{2z^2-3z+4}$$

Find the difference equation and frequency response of the system.

Question 4:



COMSATS UNIVERSITY ISLAMABAD

Lahore Campus: Defence Road, Off Raiwind Road, Lahore

UAN: +92-42-111-001-007 Ext: 810, 866

We want to design a causal discrete time LTI system with the property if the input is $x(n) = \left(\frac{1}{3}\right)^n u(n) - \frac{1}{5} \left(\frac{1}{3}\right)^{n-1} u(n-1)$ then the output $y(n) = \left(\frac{1}{2}\right)^n u(n)$. Determine the transfer function $H(z)$, the impulse response $h(n)$ and frequency response $H(\omega)$ of the system that satisfies this condition. Also assess the stability.